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A Simple Measure Of Liquidity, With Estimates From India's National Stock Exchange*

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Abstract

Market liquidity is viewed as the ease of finding a counter-party. This is determined both by price impact if an order is to be executed immediately, and by the willingness to wait and bear price risk. We motivate and define a simple empirical measure of liquidity that focuses on this attribute, by using information from both executed and unexecuted orders. We explore its implications for empirical work by estimating this measure using data from India's National Stock Exchange (NSE), and a benchmark dataset from NYSE in the US. We find that the liquidity of NSE stocks is decreasing in both price impact and waiting time, but the possibility of waiting significantly reduces the adverse effect of price impact.

March 2016 (Being revised)

* This paper has benefited immensely from detailed discussions with, or comments or suggestions from, Jordi Caballe, Damini Gupta, Craig Holden, Abhay Kaushik, Palani Krishnan, Shashi Murthy, Monali Phukan, Srini Rangan, G. Sabarinathan, Arup Mukherjee (NSE), Parvati Palaniappan, Steven Poser (NYSE), Rinell Randolph (SEC), Ashok Srinivasan and Avanidhar Subramanyam, and participants in seminars at IIM Bangalore and IIT Madras. We also thank the National Stock Exchange of India and the Securities and Exchanges Board of India, and the Securities and Exchanges Commission in the US, for providing the data that has made this research possible.

A Simple Measure Of Liquidity, With Estimates From India's National Stock Exchange

"Liquidity, like pornography, is easily recognized but not so easily defined …" O'Hara (1995, p.205)

"Keynes once observed that while most of us could surely agree that Queen Victoria was a happier woman but a less successful monarch than Queen Elizabeth I, we would be hard put to restate that notion in precise mathematical terms. Keynes' observation could apply with equal force to the notion of market liquidity. The T-bond Futures pit at the Chicago Board of Trade is surely more liquid than the local market for residential housing. But how much more?" Grossman-Miller (1988, p.617)

1. Introduction

In contrast to the above quotations, this paper believes that it is possible to define precisely a meaningful liquidity measure, derived from a primitive notion of liquidity that reconciles the many different measures of liquidity that have been used over time. We view liquidity as the ease of finding an acceptable counter-party. This leads us to define τ, an empirical measure of liquidity for a period using data from both executed and unexecuted orders: the ratio of quantities in orders that are executed, to total quantities in all orders received by an exchange. This ratio is one estimate of the probability of order execution. We explore the implications of this measure of liquidity, the executed to received ratio (or fill rate adjusted for cancellations), after estimating it using data from India's National Stock Exchange, and a benchmark US data set.

Consider a simple comparison between three alternative market scenarios faced by a potential seller who wants to sell a block of shares at a desired price of ₹525 per share, and can wait a little bit.

Scenario A: The seller can sell his block at the prevailing price of ₹500 per share. Waiting a bit does not promise to change this for better or worse.

Scenario B: The seller expects that to sell the block immediately he must accept a price of ₹475 per share. But if he waits, he feels there is a chance of finding a counter-party who would be willing to buy at the price that he desires.

Scenario C: The seller again expects that to sell the block immediately he must accept a price of ₹475 per share. And further he feels that if he waits, there is a chance he may lose even the current opportunity of a sale price of ₹475 per share, and be left facing an even lower price.

From the perspective of this prospective seller, which market is most liquid? Since liquidity is the ease with which a trader can find an acceptable counter-party, if the seller wants a price of ₹525, relative to Scenario A, Scenario B is more liquid, and Scenario C, less liquid, given the stated chances of finding an acceptable counter-party in each case. This amounts to defining liquidity over the entire period (i.e. including the interval for which the seller is willing to wait) rather than simply at a point in time.

The dominant convention in the literature has been to use a related definition, even when starting a discussion of liquidity with a reference to the ease of finding counter-parties. For example, Foucault et al (2013, p.2) state: "*Liquidity* is the degree to which an order can be executed within a short time frame at a price close to security's consensus value. Conversely, a price that deviates substantially from this consensus value indicates illiquidity: in an illiquid market, buy orders tend to push transaction prices up, while sell orders tend to do the opposite; in extreme cases the deviation is so great that it is not worthwhile or feasible to trade at all, and the market freezes." By this definition, in the above example, Scenario A in which the seller's block of shares can be absorbed immediately at the prevailing price, would be considered most liquid.

It is important to understand the implications of the definition of liquidity that focuses on the ease of finding a counter-party, and the conventional definition, and the relationship between them. The conventional definition is clearly plausible, even compelling, as a description of two *limits*, one representing a very liquid market, in which the costs of finding an acceptable counter-party in terms of both price and waiting time, are negligible, and the other, a completely illiquid market. But away from these limits, the costs in terms of both price and waiting time are not small, and there is in general the possibility of a tradeoff between price impact (often described today as the 'cost of immediacy') and waiting. Demsetz (1968) and Grossman and Miller (1988) analyze such tradeoffs, as does the growing literature based on markets organized as electronic limit order books (see Foucault, Kadan and Kandel (2005) and references therein for examples of analytical models relating to limit order books, Biais, Hillion and Spatt (1995) for an analysis of the limit order book data from the Paris Bourse, and Cohen, Maier, Schwartz and Whitcomb (1981) and Cho and Nelling (2000) for examples of empirical work that focus on the probability of execution).

In the limiting cases described in the conventional definition of liquidity it is easy to see the relationship between this definition and the definition that focuses on the ease of finding a counter-party; e.g. in a liquid market where there are negligible costs in terms of price and waiting time, it is easy to find a counter-party. Away from this limit, when there is a trade-off between price impact and waiting (i.e. in all cases other than that of a full-blown financial crisis) the value of the conventional focus only on price impact in measuring liquidity is arguable. At best, it is a measure of liquidity at a point in time, since the potential risks and benefits of waiting are ignored.

The idea relating to liquidity that is useful both at the limits and away from them is the idea of a balance between buyers and sellers. When all agents can find counter-parties easily, as in the limiting case of perfect liquidity, there is perfect balance between buyers and sellers. As costs in terms of price and waiting time rise, this balance can be disturbed, and how liquid a market is will depend on how much of a balance there is between buyers and sellers. So in this paper we treat the ease of finding a counter-party as the more primitive idea in defining liquidity, and we treat price impact or the cost of immediacy as one key determinant of liquidity. It also leads naturally to a consideration of waiting time as another key determinant of liquidity.

The limit described in the conventional definition also assumes a limit along another dimension, besides price impact and waiting time: agents' measures of the price at which they are willing to trade (arising from a combination of preferences, opportunities, and beliefs) are aligned and we have a "consensus value." Not only can a large quantity be bought or sold at the going price, implicitly it is assumed in that limit case that all participants agree on what that price should be, ruling out a case like the ones we considered in the introductory market scenarios A, B and C, where the seller seeks a price of ₹525 when the going price is ₹500 per share. If, as is usually the case, we do not have a consensus price, it is not clear how we would go about measuring whether a market is more liquid or less liquid with the conventional definition. The empirical practice has been to focus on the cost of immediacy and implicitly assume everything else is constant. In the opening example of this paper, in Scenario A the market can absorb the seller's block without any deviation in price, but the seller wants an even better price. From his perspective, Scenario B promises a greater likelihood of finding an acceptable counter-party, and so we would consider it more liquid. Agents do not come to the market to discover the right price, or the true price. They come only to find a counter-party, and observing a price is only incidental to deciding if a counter-party is acceptable. So it is important to be able to define liquidity without relying on the price, or on any degree of consensus over what the price should be.

When agents' beliefs or more generally, the price at which they are willing to trade, varies (i.e. we do not have price discovery in the sense of a consensus price), focusing on the ease of finding a counter-party can still enable us to rank-order different market protocols in terms of liquidity. Suppose we have four agents who have submitted limit orders in the market, a buyer-seller pair with a high valuation (say, ₹10,000) and another buyer-seller pair with a low valuation (say, ₹1,000). If the high-valuation seller is matched with the with the high-valuation buyer, and similarly, if the agents with low valuations are matched with each other, then all agents would find acceptable counter-parties, and the market would be liquid. Whether this will happen in practice can vary depending on market protocol. In an open electronic limit order book, this will depend on the order in which the four agents submit their orders. In a batch call auction, the volume maximizing price is some price in (₹1,000, ₹10,000), and only the high-valuation buyer's and the low-valuation seller's limit orders would be executed. And it is conceivable that we could design some other market protocol that could match buyers and sellers better. The practical point is that by focusing on the degree to which buyers and sellers are matched we can define a measure of liquidity, independent of any assumption about price discovery or consensus over price.

So in this paper, we propose our empirical measure of liquidity for a period, τ: the ratio of quantities in all orders that are executed,[i] to quantities in all orders available for execution, i.e. all orders received, in a period. The focus on liquidity in a period, rather than at a point in time, reflects the belief that for most final customers, that is what matters. It is also necessary to define liquidity over a period to account for the role of an agent's willingness to wait.

So our empirical measure of liquidity uses information from both executed and unexecuted orders. Given the limitations of our dataset from India's National Stock Exchange, the smallest period we can consider is the trading day. In a perfectly liquid market where all agents find counter-parties, τ approaches one. In the opposite extreme, where no orders are executed, τ is zero. Notice that we do not rely on price information to define this liquidity measure.

We present measures of τ for stocks listed on India's National Stock Exchange. We then examine the relationship between this liquidity measure τ and its principal determinants, price impact and waiting time. The interaction term between these two principal determinants also tells us about the price benefits or risks of waiting. To determine whether our liquidity measure would make a difference, relative to a focus on price impact, we examine empirically the total derivative of our liquidity measure with respect to price impact. Our results show that while τ is declining in price impact and waiting time, the coefficient of the interaction term is always significantly positive. And the direct reduction in liquidity caused by an increase in price impact can, for many thinly traded stocks, even be swamped by the indirect effect of waiting.

This paper is organized as follows. Section 2 describes the relationship between our work and previous work. Section 3 describes the data from India's National Stock Exchange (NSE) and constructed variables. Section 4 presents estimates of liquidity on NSE, and analyzes its determinants such as price and waiting time, which are relevant in all markets, together with variables such as the existence of single-stock futures markets and Foreign Institutional Investor (FII) investment, which are considered important in India. Section 5 provides a limited

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[i] Adjusted for subsequent cancellations by the order submitter herself.

comparison of NSE liquidity with an available benchmark from the US: NYSE stock liquidity as gleaned from SEC Rule 605 compliance filings. Section 6 analyzes liquidity and returns simultaneously. Section 7 concludes.

2. Relationship To Previous Work

This paper takes liquidity to be the ease of finding a counter-party, and our empirical measure of liquidity τ stems directly from this. The concept of market liquidity has been defined in many different ways in previous work. Amihud (2002, p. 33) notes: "Liquidity is an elusive concept. It is not observed directly but rather has a number of aspects that cannot be captured in a single measure." O'Hara (1995, p.205) has even suggested that "liquidity, like pornography, is easily recognized but not so easily defined …" We believe this reflects how for the extremes of perfect liquidity or complete illiquidity, regardless of one's liquidity definition, one would generally reach the same conclusions for whether a market is liquid or illiquid, a point also made in Grossman and Miller (1988). But as noted in our introduction, for all the cases in between the extremes, intuition from the limit cases is not useful.

To Black (1971) and Kyle (1985), a liquid market exhibits *continuity*, i.e. small orders have small price consequences, and *efficiency*, even in the face of private information, so large orders have large consequences. So a liquid market is a market in which bid-ask spreads are always available, and small, and where larger blocks should be expected to move prices more if immediate execution is demanded, and very little if execution can be spread over a long period. Black (1971) also explains lucidly why price continuity is not necessarily a desirable property and why a liquid market should *not* be infinitely deep. He even states explicitly: "The ability to handle large amounts of stock in short periods of time without changing the price of the stock is *not* [emphasis added] a characteristic of a liquid market." Thus the Kyle (1985) model is a model of a liquid market, in the Black-Kyle sense of small orders having only small price impacts. Every Kyle model is also a liquid market in our sense, with waiting time being zero: every agent finds a counter-party and all orders are always executed, by market makers who stand ready to serve as counter-parties with unlimited amounts of cash and stock. So the vast literature following Kyle (1985) has tended to focus on price impact, Kyle's measure of market depth, $1/\lambda$, itself as a measure of liquidity. Our data comes from a market where, as in most real-world markets, all orders are clearly not executed, and so we assume a setting where both price impact and the willingness to wait and bear price risk affect the likelihood of finding a counter-party in a given period.

Black (1971, pp.29-30) also refers in passing to what we take as the premise of this paper: the idea of finding a counter-party as underlying one meaning of liquidity. This notion has been referred to in papers such as Amihud et al (2005, p. 270) who call it "the ease of trading." O'Hara (2002, p.1338) also writes: "Liquidity refers to the matching of buyers and sellers. It is intertemporal in nature and it is not necessarily linked to price discovery." But this idea of finding a counter-party has not generally been made the basis of an empirical measure of liquidity.

The closest we have found to this is in the literature that uses the level of trading volume or trading frequency as an indicator of its liquidity. Fleming (2003) evaluated a trading frequency measure as a measure of liquidity, and found that it performed poorly when he took the bid-ask spread as a benchmark measure of liquidity. As no trading implies no change in price, Lesmond, Ogden and Trzcinka (1999) invert this relationship and assume zero-return days reflect no-trade days, and the fraction of no-trade days is their measure of illiquidity. Foucault et al (2013) suggest that volume can reflect information and so these measures are problematic as measures of liquidity. These measures also do not distinguish between zero volume because no orders are submitted and a zero volume because no submitted order is executed. Given a large number of thinly traded stocks, this distinction could matter. Our measure differs from these in using information from both executed and unexecuted orders directly, and in not relying on returns or prices in any way.

Even authors who have noted that liquidity is related to the idea of finding a counter-party (e.g. Amihud et al (2005, pp.270-271)) move on quickly, like Foucault et al (2013, pp.4-5), to discussing illiquidity, the cost of illiquidity, and compensation investors seek because of illiquidity. Illiquidity and transaction costs are sometimes used by these authors as synonyms. This focus on the problem of investors and portfolio managers, and the relative paucity till recently of detailed data from limit order books, has led naturally to studying liquidity measures defined in terms of the price impact, which we view not as liquidity itself but as a key determinant of liquidity. This is clearly important work, which investors and portfolio managers concerned about the cost of capital care about deeply.

Over time there have been a large variety of liquidity measures based on price impact that have been used (e.g. by using Kyle's lambda for order flows and price changes accumulated over fixed periods, the bid-ask spread, or related ideas such as the effective spread, Amihud liquidity, Roll's measure, and the Probability of Informed Trading (PIN), which given models like Glosten and Milgrom (1985) is monotonically related to the bid-ask spread). These price impact measures differ in the data that they use (microstructure detail or just returns, or returns and volume), the periodicity with which the measures are computed (5-minute intervals for some measures of Kyle's lambda, or daily or less frequently for return-based measures), the choice of benchmark against which performance has been measured, and so on.

Even for the cases where our regression results suggest that an increase in price impact need not lead to a decrease in liquidity, because the indirect effect on liquidity through waiting time swamps the direct effect of a price impact, our results do not reduce the value of the existing literature on pricing the liquidity factor for investors and portfolio managers. Exactly what the label for that factor should be – whether it should be called liquidity as those papers do, or as a determinant of liquidity, as we suggest, or even something completely arbitrary, simply does not matter. At its core that literature has identified a factor that can be measured using a well-specified algorithm, and that factor is priced by investors. That alone is relevant. It is possible of course that whether or not the computed measure of liquidity varies with the ease of finding a counter-party can explain some of the cross-sectional or time series variation in what that literature has called the liquidity coefficient.

Does this mean that how we measure liquidity does not matter? As many authors including Grossman and Miller (1988), O'Hara (2002), Foucault et al (2013) have noted, measuring liquidity is still important. Our liquidity measure τ is best seen as a measure of market performance or market health. A market with a better balance between buyers and sellers, where a greater proportion of orders is executed is healthier than a market where a smaller proportion of orders is executed.

Regulators, market designers, and even investors and potential market participants, all can benefit from that information. For example, if our measure τ is relatively high, suggesting a balance between buyers and sellers, whether this is because the cost of immediacy (price impact) and the need to wait for a better counter-party are low, or because the cost of immediacy is high but agents are able and willing to benefit by waiting a bit, or whether τ is high in spite of both the cost of immediacy and the need to wait for a better counter-party being high, tells us a lot about the nature of the market and its participants. This information can help agents – be they investors, regulators, market designers – make better decisions. While the popular press often interprets mere movements in prices or indices as indicative of market sentiment, it is clearly more informative to know whether such movements are accompanied by significant volume, and by adequate balance between buyers and sellers.

What about the vast literature that describes various determinants of the bid-ask spread and other measures of price impact, such as exogenous transaction costs, demand pressure, inventory risks, asymmetric information, and so on? Those factors are obviously still important. Only rather than regarding them as direct determinants of liquidity, we would count them as determinants of liquidity through their influence on the direct determinants of liquidity such as the price impact and waiting time.

To sum up, the idea of liquidity as the ease of finding counter-parties is not new. It has been referred to at least in passing in prior work. Our real departure from the literature is that we invoke this as a primitive idea to define even the empirical measure of liquidity, τ, the ratio of quantities in executed orders, to quantities in all orders received. And we look at the influence of not just the price impact but also of waiting, on liquidity.

In what follows it is important to note that we are not seeking to model or estimate the impact of an individual intra-day transaction. We view Grossman and

Miller (1988) not as a model for an individual intra-day transaction but as a summary of what happens over the course of an entire trading day.

3. Data and Constructed Variables

The Indian setting is unique in several respects. India is classified as an emerging economy. Yet unlike many such economies it has a long history of functioning democracy, and a significant institutional framework within which its economy operates. Since 1991 substantial licensing requirements restricting the scope of private sector activity and many foreign currency controls were eliminated in India. There has been a steady growth in the economy, with a consequent increase in the degree to which the stock market has been tapped for funds by both Indian and international investors. India's regulatory framework provides that securities should be traded under the supervision of the stock exchanges and the Securities Exchange Board of India (SEBI), India's equivalent to the US Securities and Exchanges Commission (SEC). The number of stocks listed on India's stock markets is very large (though active trading is concentrated, as in most markets of the world, in only a few stocks).

The oldest amongst these stock exchanges in India is the Bombay Stock Exchange (BSE). It was set up in 1875 and at one time was the principal exchange in the country, accounting for more than 90% of all stock market activity till the mid-90s. This was largely a middleman controlled exchange, though in May 2007 a move towards demutualization took place. Today BSE has lost substantial ground to the newer National Stock Exchange (NSE) which was promoted by several financial institutions.

NSE began operations in 1994, and organized its trading in a manner so as to increase its informational efficiency and provide market transparency. It represents an order-driven market like the Paris Bourse. Its trading system is known as NEAT (National Exchange for Automated Trading) and is an online, fully automated, screen based trading system where trades are matched electronically and executed on a strict price-time priority basis. While BSE has more stocks listed on it than NSE, and is larger in terms of market cap, NSE is several times larger than BSE in terms of the value of equity shares traded, and also accounts for more than 90% of all single stock futures trading. Most active stocks in India are listed on both BSE and NSE, and many brokers are members of both exchanges. In terms of the average daily number of trades, NSE has ranked third in the world in several recent years; in the average daily dollar-volume, it has been ranked between eleventh and fifteenth in different years.

Our NSE sample data for equity stocks extends from January 1, 1999 to March 31, 2011.[ii] NSE provides snapshots of the order book (i.e. a listing of all unexecuted orders) for most days at four time points during the day, 11am, noon, 1pm and 2pm. This data enables us to compute

(i) quantity in unexecuted orders at each snapshot; to best approximate the end of day unexecuted quantity, we used the last snapshot;

(ii) impact-cost, which is computed per the definition used in many exchanges round the world that operate as electronic limit order books. To compute impact cost, we use the orderbook snapshot files for each equity stock in our sample. To compute impact cost we first calculate, for each firm i, snapshot s, and $d = B\ (\text{buy}), S\ (\text{sell})$, the percentage difference between the weighted average execution price and the pre-trade mid point, for an order size of ₹100,000. It is given by

$$Z_{i,s,d} = \frac{\sum_{1}^{\bar{k}} p_k q_k - P_m Q}{P_m Q} \text{x} 100 \qquad (1)$$

where P_m = the average of the best bid price and of the best ask price; the index k runs across orders on the opposite side (i.e. buy orders if we are computing the sell-side impact cost) arranged in price-time priority, with the lowest index representing the best order; and $\bar{k}$ is the smallest index such that $\sum p_k q_k$ just equals ₹100,000 (and $q_{\bar{k}}$ is chosen from the available quantity on the $\bar{k}^{\text{th}}$ order to

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[ii] Because we anticipate receiving data from NSE that goes beyond order book snapshots to detailed lists of orders submitted and orders executed, the following empirical measure and discussion will be replaced by a more direct measure of the fill rate (adjusted for cancellations by the order submitter). But that data will come only from more recent periods. This data will also allow us to measure time to execution directly.

make $\sum p_k q_k$ equal to ₹100,000); and $\sum q_k = Q$. If, however, available orders are insufficient to meet this standard order size, then $\bar{k}$ represents the highest index. The percentage impact cost for each firm i, snapshot s, and $d = B \text{ (buy)}, S \text{ (sell)}$, is given by

$$IC_{i,s,d} = \begin{cases} Z_{i,s,d}, \text{ if the order size of rupees 100,000 can be met from available orders,} \\ Z_{i,s,d} + 5\% * (100000 - Z_{i,s,d}), \text{ otherwise.} \end{cases}$$

The order size of ₹100,000 is arbitrary and we use it only because SEBI now requires that exchanges use this number in computing impact costs. The same SEBI circular also requires that exchanges assume any shortfall from the standard order size of ₹100,000 will be met at an impact cost of 5%. Shah and Thomas (1998) mention that this 5% is an estimate of the higher cost of meeting a shortfall from any of the 22 other exchanges in India. Finally we take a simple average of the buy impact cost and the sell impact cost.

(iii) age of orders, by subtracting the snapshot date-time from the order submission date-time, measured in hours. We have examined several kinds of age variables – age of the best, newest, oldest order, or the average age of orders within a snapshot for a firm – because they are all highly correlated and our results hardly vary, we only report results based on age of the best order.

To capture the notion of waiting time, ideally we would have liked to measure the time to execution for each executed order, and the age of each unexecuted order. Our current data does not allow us to do that. Since the execution lag must in general be correlated with the age of unexecuted orders, we use information about the latter in order book snapshots as a proxy for the former. All snapshots during a day were used, to get a measure of average waiting time per firm-day.

In recent years in the US, agents who do not insist on immediacy, often do not wait with their orders till the end of the day, but cancel an order and place a fresh one. This strategy has reportedly increased with the growing reliance on particular kinds of algorithmic trades. In any case it is important to note that order cancellations only imply that the maximum age of an order is likely to be smaller. It will still be the case that the age of unexecuted orders is positively correlated with the age of executed orders, i.e. execution time.

It is also important to note that while zero waiting time and an arbitrarily large waiting time will be associated with immediate execution and non-execution, away from these two limits (and that middle is where all of the empirical data comes from) the average age of orders and execution probability during the day are not tautologically equivalent. An observation with a higher average age of orders than another could have either a higher or lower τ for the entire day, depending on whether waiting increases or decreases the likelihood of a counter-party emerging. Similarly, even if algorithmic trading strategies are similar to those used in the US to learn about the state of the market by using automatic small orders and cancellations to "ping" or probe for volume on the other side, a daily observation reflecting more cancellations during the day may yet yield a higher or lower τ or execution probability for the entire day, depending on whether or not such algorithmic strategies actually help agents learn how to find a better match.

NSE also provides "bhavcopy" files, which are daily summaries. From these we compute

(iv) daily trading volume,

(v) daily returns for each firm.

For some of our analyses, we relate market liquidity not only to its primary universal determinants, price impact and waiting time, but also to other determinants that are of special significance to India. India is one country where single-stock futures markets have significant volumes (in fact usually larger than the underlying equity volumes, since settlement with cash rather than delivery is not only permitted but often used). Single-stock futures trading in India has been permitted case-by-case in some of the most active stocks. So whether or not such futures exist is an indicator of whether a firm is large and well-established. This data comes from NSE.

Futures trading in India is almost exclusively concentrated in the National Stock Exchange. Trading began in index futures contracts in June 2001, and in single-stock futures contracts in November 2001. In contrast in the US, single-stock futures were banned in 1981, and when restarted in 2001, the two exchanges that launched single-stock futures (OneChicago and NSLQ) found that volume dwindled quickly, and both have now closed down. The stated argument for banning single-stock futures trading in 1981 was that manipulation in single-stock futures was considered too big a risk, despite possible gains to price discovery from the activities of informed speculators. So futures trading was permitted in the US (and in many other countries) only in index futures. Our NSE single-stock futures data starts from February 2002, and runs till March 2011.

Also Foreign Institutional Investors (FIIs) are generally believed to play a significant role in the Indian economy since its significant opening up in the early 90s. We obtain data on FII transaction records from the Indian regulator, Securities and Exchanges Board of India (SEBI), India's counterpart to SEC in the US. This lets us compute the daily FII purchases and sales (quantity and rupee value) for each firm, and we construct some related variables from these. FII data is available from the time that daily electronic filing of FII transactions with SEBI by foreign exchange custodians was mandated in late 2001, until December 2012. We use the data from 2002. Before 2002 when there is no FII data we cannot tell whether this is because there was no FII activity at all, or whether it was only that such activity was not reported, since reporting was not required. In unreported work we experimented with several different FII variables, many of which are strongly correlated. The regression analyses in this paper use the daily net cash inflow due to FIIs, scaled by the daily trading value for the firm.

The basic focus in this paper is on our measure of liquidity, τ, the ratio of quantities in orders that are executed, to quantities in all orders received within the period. It uses information from both executed and unexecuted orders.

τ = (Trading volume for the period)/(Trading volume for the period + Quantity in unexecuted orders during the period), or, more specifically, given the limitations of our data,

$$\tau_{\mathrm{it}} = \frac{(\text{Trading Volume on the NSE})_{\mathrm{it}}}{\{(\text{Trading Volume on the NSE})_{\mathrm{it}} + (\text{unexecuted\_order\_quantity}_{\mathrm{it}})\}}$$

where

τ_{it} = the Effective Liquidity Score for Firm i on Day t,

$\text{unexecuted\_order\_quantity}_{it}$ = sum of quantities in buy orders plus the sum of quantities in sell orders, for Firm i, on Day t, in the last order book snapshot on day t.

The quantity in orders executed during the day – the daily trading volume – is reported in the NSE bhavcopy files, so there is no need for estimation or approximation. We do not however have data that lets us determine exactly the quantity in unexecuted orders. So we need to estimate unexecuted quantity for the day from order book snapshots, available usually for four time-points during the day. We have experimented with different estimates of unexecuted quantity from order book snapshot data, and they are all highly correlated. For purposes of our subsequent regression analyses, it makes little difference which particular alternative we use.

We finally picked the unexecuted quantity from just the last snapshot of each day. This number will understate the actual unexecuted quantity to the extent that there are order cancellations earlier in the day. We judged this error to be small because in India computer-based trading algorithms which generate a lot of the cancellations in US markets are still rare.[iii] So an unexecuted order in an earlier snapshot is most likely executed later in the day, or remains as part of a later snapshot. Conversely, the number in the last snapshot will overstate the unexecuted quantity for the day, to the extent that an order listed in the snapshot is executed after the snapshot time but before the end of the day. Using just the last snapshot makes this error smaller than with an alternative that also uses earlier snapshots. Given current data limitations, we feel this is the best approximation possible. As the exchange makes more detailed order book data available, we could of course refine this measure.

Table 1 provides some descriptive statistics of the primary variables in our study. The data reveals a lot of variation on the NSE. Panel A describes the variation in the overall sample. τ ranges from zero to one, so we have firm-days where no orders are executed and where all orders are executed. The median firm-day shows only 37% execution, while the mean is about 40%. Waiting time proxies

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[iii] This has changed in recent years where algorithmic trading has also increased in India. The new data will help avoid approximating cancellations by giving us that information directly.

(age of the best order, average age of orders) and impact cost also show a lot of variation. The minimum age of the best order is about .03% of an hour, which is about 1.8 seconds. The maximum age of the best order is 1180+ hours, which is over 49 days. This is because till 2004 pending orders did not automatically expire at the end of the day, so some stale orders were carried in the books.[iv] Since then, following the practice in most western exchanges, NSE also requires pending orders to expire at the end of the day. However, even earlier most orders were very short-lived, as is shown by the 99th percentile being dramatically smaller than the maximum age. Impact cost varies from less than 1/10 of 1% to over 200%. Note that the mean and median price impact over the entire sample of firm-days are about 2.4% and 1% respectively, the 90th percentile is 6%, and the 99th percentile, 13%. In the subsequent regressions, we winsorize the extreme 1% of all variables used in the analysis. Only about 10% of all firm-days have single-stock futures. And FII activity ranges from accounting for all the selling to all of the buying on a given firm-day. Panel B, which summarizes the variation in firm-by-firm averages of the same variables as in Panel A, also confirms that firms are very different. It also shows that the firm-wise average net daily FII cash inflow, scaled by daily trading value, over this period has been non-negative.

4. NSE Market Liquidity and its Determinants

We first compute τ for NSE at the aggregate level, aggregating across all stocks, and for quintiles of firms formed by average daily trading volume, for each trading day, for our sample of NSE stocks. This is shown in Panel A of Table 2. The overall mean and median are 40% and 38%. There is a sharp falloff in liquidity as a stock is more thinly traded, with mean and median τ dropping from 63% and 68% in the most active quintile to 21% and 15% in the least active quintile. In Table 2, Panel B, we provide some idea of the variation over time. There have been cycles with liquidity low points which bear some relation to significant financial crises, as in

[iv] We have also dropped entirely the pre-2004 data, with little change in qualitative conclusions. In the revision we will be focusing exclusively on more recent data, because individual order books that start and end each day at zero are also less susceptible to endogeneity problems.

2001, with the big dotcom bust, and 2008, when the current financial crisis reached a tipping point.

Under the perspective adopted in this paper we view price impact or the cost of immediacy as a key determinant of liquidity, and waiting time as a second key determinant. The basic regressions we report on regress τ on these two key determinants, and an interactive term, which captures the price risks and benefits to liquidity of waiting. We also explore the impact of one control variable, market returns, intended to capture overall market sentiment.

These determinants matter everywhere. Given that our data set is from India, we also explore the impact of whether single-stock futures exist for a given firm, and of FII activity.

Our results are summarized in Table 3, in which we report results from 4 different regression models, which implement GMM estimation with Newey-West errors that allow for serial correlation and cross-sectional heteroskedasticity. Models 1 through 4 start from the basic price impact and waiting time variables, and sequentially add the futures dummy (=1 if single-stock futures exist for that firm), net FII cash flow scaled by the daily value of shares traded in a firm, and market return. We have also run additional models that use log-transformed, rank-transformed variables for all but the futures dummy F, and quantile regressions, with little difference in key qualitative results.

The results are remarkably stable. First note that adjusted R-squared is 34% for the most parsimonious model, and rises to 41% in other cases, which seems high relative to customary results in financial markets using daily data. The direct response of τ to a change in price impact and waiting time is significantly negative. For each additional percentage point of price impact, liquidity as measured by τ declines by over 5 percentage points. For each additional hour that an order remains unexecuted, τ declines by over 3 percentage points. But the interaction term between the price impact and waiting time is always significantly positive; i.e. it appears that by not insisting on immediacy market participants are able to find a counter-party willing to accept a more favorable price. This suggests a setting analogous to Scenario B in the opening example of this paper best describes the markets from which we draw our data.

The sample for the analyses in Table 3 consists of a pooled (cross-section and time series) sample. We then run each of our regressions firm-by-firm, and summarize the results across all firms with cross-sectional descriptive statistics in Table 4. While the numbers for the maximum firm-specific coefficients suggest there are a handful of outliers, the broad conclusions from the median values in Table 4 are consistent with, and very similar to, those from Table 3.

The existence of single-stock futures has a big impact on liquidity in the underlying equity market, by raising τ by about 24%. It also alleviates the indirect effects on liquidity through price impact and waiting time significantly. Foreign Institutional Investors (FIIs) also have a positive and significant direct effect on liquidity, with a slightly under 0.1% increase in τ for a 10% increase in the net cash inflow due to FIIs as a proportion of the total daily value of trading. The indirect effects of FII activity are no significant. Market-adjusted firm return, which we regard as a measure of overall investor optimism or pessimism, has a significant negative impact, suggesting that when investor sentiment moves up it creates a greater imbalance between buyers and sellers.

To see whether in computing the total derivative of τ on the price impact, the indirect effect via the interaction with waiting time swamps the direct effect is of some interest. We compute this total derivative for each of the models estimated in Tables 3 and 4. Table 5 provides summaries of three different ways in which the total derivative $d\tau$/d(price impact) was analyzed. In Panel A we evaluate this total derivative at each observed value of waiting time and other variables causing interaction effects, then for each firm we determine the proportion of cases for which this total derivative was positive. It turns out to be so in about 20% of all cases. In Panel B we evaluate the total derivative at the mean values, and in Panel C, at the median values, of the variables causing interaction effects, and at least for a small but significant proportion of firms, it is clear that we cannot assume that as price impact increases liquidity declines.

The reader should note that even where the indirect effect is not large enough to swamp the direct effect of the price impact on τ it may yet be large enough to change relative liquidity rankings between firms and time periods. We computed firm-wise Pearson and Spearman correlations between τ and price impact, and then computed descriptive statistics of these correlations across firms. While the mean and median correlations are significantly negative, confirming again that for most firms as price impact increases liquidity declines, the mean and median values are only about -0.2, and for about a fifth of the firms the correlations become either insignificant or even significantly positive. So while there is little doubt that price impact is a key determinant of liquidity, other factors are also important.

5. Comparison of India's NSE with NYSE in the US

Is liquidity on NSE high or low? This cannot be answered in the abstract, and requires comparison to a benchmark. To compare India's NSE with a benchmark from a developed market, we use US data from the New York Stock Exchange (NYSE), created to comply with SEC Rule 605 ("Disclosure of Order Execution Information") under Regulation NMS, a 'series of initiatives designed to modernize and strengthen the national market system ("NMS") for equity securities.' This data from the NYSE reflects some unique features and constraints. Unlike India's NSE which (like some other exchanges like the Paris Bourse) discourages off-exchange trading, and requires all orders to pass through the central limit order book,[v] the NYSE permits off-exchange trading and anecdotal evidence suggests that on some recent trading days over 40% of all volume has been generated off-exchange.

Some of these off-exchange market centers, referred to colorfully in the US as dark liquidity pools, provide much less transparency. The only specific order related information that is made available is if it results in a trade, when ex-post the trade is added to the NYSE Consolidated Tape. These off-exchange market centers are also required to report aggregate information about order execution under SEC Rule 605, albeit under participant codes distinct from the code under which NYSE reports these statistics for its own exchange-based activity. Till 2005 the SEC consolidated filings from all market centers and made the consolidated 2001-2005 data widely available. But since then the SEC only requires market

[v] There is a separate book at NSE for block trades from other equity trades, though all of this is observable to all market participants. In its early days NSE also had a separate book for negotiated trades which reflected off-exchange trades. For at least about a decade now, this book has ceased to exist, and the exchange claims to be strict in its prohibition of off-exchange activity.

centers to maintain publicly accessible records on their own websites for three months, and most choose to do no more than that.

The NYSE itself though has publicly accessible records going back to 2001, which allows a longer series for comparison, and that is what we use, for the US benchmark. Rule 605 data is aggregated by month, so we can estimate τ only for the month. Because in the US it has long been the rule that pending orders expire at the end of the day, this monthly τ is also equivalent to the monthly average of daily τ numbers. In India this became a rule only in 2004 but by also computing monthly averages of NSE daily τ numbers, we get more comparable numbers.

To compute the NYSE τ numbers from Rule 605 data we define the denominator as quantities in all orders initiated on NYSE, and compute the numerator as quantities in all orders initiated on NYSE, and executed at NYSE or elsewhere. These numbers do not include orders initiated off-exchange. Nor do they include orders initiated away from the NYSE but executed at NYSE. Given the limitations of the data, and our desire to have as long a comparison series as possible, this is the optimal US benchmark we could define.

We compare the monthly time series τ at the aggregate level from June 2001 till March 2011. We have 2208 firms in our NSE sample, and 7235 firms in our NYSE sample, for which we had comparable data over this period. Pooling across all stocks on the exchange means that the stocks that contribute the largest quantities in orders received, and in orders executed, receive more weight in the computation for τ than the stocks with more limited activity in them. Figure 1 presents a time series comparison between NSE and NYSE. At the start of the decade, NYSE numbers show much higher liquidity than NSE (almost 50% to 27%). But the salient feature of this graph is the generally declining trend in the NYSE τ and the generally increasing trend in the NSE τ. In recent months the execution probability on the NYSE has fallen to below 10% (and per data on the SEC's website, even below 5% since our sample comparison period which ends in March 2011). NSE numbers have sometimes exceeded 60%, and while generally lower since February 2010, have stayed over 40%.

Recall that for NSE we use an estimate for unexecuted quantity, and for NYSE the numbers exclude significant off-exchange activity. So a comparison of raw liquidity numbers must be interpreted cautiously. But the differences in trends over time seem more robust. In evaluating the comparison between NSE and NYSE, beyond differences between a developing and a developed economy, and any broad differences arising from history and geography, the two exchanges differ with respect to the existence of and the attitude towards off-exchange activity. NSE prohibits off-exchange activity, and there appears to be little. NYSE permits off-exchange activity and especially since 2006 there is a lot.

A conjecture for sharply declining τ on NYSE is the exponential growth in the use of algorithmic trading. Over time the number of orders has increased but the average size of orders has declined. This is believed by some to reflect computer-based strategies that seek to "ping" or probe the other side for volume, in order to learn more about the state of the market. While there are anecdotal reports that some participants such as Goldman Sachs have benefited enormously from such algorithmic strategies, our results raise questions about whether the market as a whole has been learning more about counter-parties from such strategies.

A market participant has a better chance that small pinging orders will uncover a large "iceberg" order with significant hidden quantity if he is the only person using such a strategy. In that case even if the pinging strategy involves cancellations of many small orders, we should also subsequently see significant iceberg quantities being executed. But when the whole market uses such a strategy, small pinging orders are more likely to bump into each other and be less likely to uncover icebergs. And a setting where most players use frequent cancellations makes it harder to develop a reliable measure of supply or demand, and to find counter-parties. Whether US markets reflect a setting where pinging strategies are now laden with significant negative externalities is an important question for future research.

In India reliance on computer-based trades seems limited (for the period considered in this version). Even less clear is the composition of algorithmic trading strategies. If algorithms are used only to constantly identify and exploit small arbitrage opportunities, we would expect them to improve the probability of order execution, unlike in the case of pinging strategies where the effects are ambiguous. Chakraborty and Balwani (2013) note that US experience with volatility arising from algorithmic trading has led to rules round the world creating disincentives for some trading algorithms. Exchanges in India levy a penalty on trading firms that place a large number of orders that do not result in actual transactions, and the regulator SEBI recently required them to double that penalty. Algorithmic systems are also required to be audited every six months.

But we note that some "flash crashes" as in the US have been reported in India in very recent periods (after our sample period) – see, e.g. Shenoy (2012) and Shaaw, Chakraborty and Balwani (2012). We also note that a google search using as keywords "India algorithmic trading" and "India program trading" does reveal several Indian brokerage websites offering access to algorithmic trading systems, and these commercial sites understandably hawk their benefits. Quantitative data on the extent to which these are actually used was unavailable till very recently, and even in public reports by the exchanges and the regulator we do not have any indication of the extent and composition of algorithmic trades in India during our sample period. Only recently have algorithms been required to use a special code when submitting an order, which can help in monitoring compliance with rules.

We drill a little deeper into the differences between NSE and NYSE. All exchanges are known to have activity concentrated in a small number of stocks. To get some insight into relative concentration and its impact on liquidity, we compute the average τ for each firm across all months. Figures 2 and 3 show respectively, the proportion of firms and the number of firms, with τ exceeding a given value. While our data shows that NYSE has more firms at both extremes, with τ = 100% and τ =0%, Figure 2 shows that for the most part the NSE graph lies well above the NYSE graph, as for almost any y, τ >= y for a higher proportion on NSE than on NYSE. But it is important to recognize NYSE reflects 7235 firms' data, while the NSE sample has only 2208 firms. Figure 3 shows that for the first 2208 firms the τ values on NYSE are systematically higher the corresponding τ values on NSE. But overall it has far more listings, and it also has many firms with very low τ values, to the point where the aggregate numbers are dominated by the large number of low τ firms.

Figures 4 and 5 show the extent of concentration more directly, by plotting cumulative volume (executed quantity) and cumulative quantity in all orders received, as a function of the proportion of firms and the number of firms. Both on NSE and NYSE there is significant concentration of activity in a few firms, and on both exchanges, the concentration in terms of quantity in orders executed is greater than the concentration in terms of quantity in all orders received. From Figure 4, about 90% of the trading volume is accounted for by just 22% of all firms on NSE, and by just 23% on NYSE. About 90% of the quantity in all orders received is accounted for by 28% of all firms on NSE, and by just 24% on NYSE. Figure 5 which examines concentration as a function of the number of firms rather than the proportion of all firms confirms that while both exchanges exhibit concentration, NYSE, the larger exchange, has more firms with both significant activity, and with very little activity, than NSE.

An SEC official has suggested that market and marketable limit orders on NYSE would show much higher τ values. Given the intended scope of this paper, our focus on analyzing India's NSE, and the lack of comparable data for NSE, we do not investigate this further. But we note in passing that if market and marketable limit orders do show much higher liquidity, our overall NYSE numbers must be dominated by limit orders which have even lower liquidity. This would suggest that market participants care much less about execution than about limiting the downside in the event of execution, an indication of greater caution in recent periods. It is possible that market participants on India's NSE are less cautious.

One comparison between India and the US in terms of liquidity appears in Fong, Holden and Trzcinka (2013). Their Figure 1 provides a comparison of the equally-weighted mean of the monthly percent effective spread from January 1996 to December 2007, for several exchanges including India's Bombay Stock Exchange (BSE) and NYSE. The percentage spread on NYSE has always been under 2% in this period and has declined over time to even well under half a percent. In many stocks the quoted NYSE spreads are now at the minimum of $0.01. BSE effective spreads have fluctuated over time, though declining over 80% in that period to end up around 3% in 2007. Because most large stocks in India are listed on both BSE and

NSE, and many brokers are members of both exchanges, we can infer that NSE spreads are roughly in the region of the BSE numbers for most stocks.

By this traditional comparison of price impact, NYSE stocks would seem to be more liquid than NSE stocks. Looking also at τ reminds us that on NYSE execution probabilities have declined sharply, so examining only price impact leaves the liquidity picture somewhat incomplete. Ceteris paribus, we would expect sharply lower price impact to lead to a much higher probability of execution. Clearly ceteris paribus does not hold. With appropriate detailed order data, one could check if this is because NYSE market participants also generally prefer, like market participants in India's NSE, not to insist on immediacy, but to wait for a better counter-party. When even virtually negligible price impact does not lead to an acceptance of immediacy, it strongly suggests that agents are holding out for more favorable prices, as in Scenario B in the introduction to this paper.

6. Returns and Liquidity

One concern in previous empirical work studying liquidity and returns has been to see the effect of liquidity on firm returns. It provides a measure of how much investors need to be compensated for liquidity risk. Because there is a substantial a priori risk of endogeneity arising from mutual causation between liquidity and firm returns, we decided to run a simultaneous equation model in which τ and market-adjusted firm returns are simultaneously determined. The other determinants are essentially the same factors as in the regressions explaining just τ in Tables 3 and 4. Because firm returns are adjusted for market returns we drop the market return as an independent variable. Also, to minimize concerns arising from endogeneity of FII activity, we include the FII variable only in lagged form.

Table 6 summarizes our results. The adjusted R-squared for the τ equation is over 60%, while for returns it is less than 4%. Our conclusions about the relationship between τ and its key determinants remain robust to simultaneous estimation with the equation for daily firm returns. We find from the liquidity equation that τ is still significantly decreasing in price impact and waiting time, but the interaction effect is still significantly positive. The existence of single-stock futures raises τ substantially, though the magnitude of the response has dropped to about 9% in Table 6 relative to about 24% in Table 3. Liquidity rises about 90 times more in contemporaneous return than it falls in lagged return. From the return equation, consistent with prior research using other measures of liquidity, market-adjusted firm returns are also significantly related to our liquidity measure τ. But it is increasing only slightly more in contemporaneous liquidity than it is decreasing in lagged liquidity. Lagged FII is insignificant. A priori it is not clear whether FIIs use up or supply liquidity, and whether FIIs create higher returns or go where returns are higher. Future research should examine the role of potential diversity in incentives across FIIs.

7. Concluding Remarks

Market liquidity is viewed as the ease of finding a counter-party. We motivate and define a simple empirical measure of liquidity, τ, that focuses directly on this attribute, by using information from both executed and unexecuted orders. We explore its implications for empirical work by estimating τ using data from India's National Stock Exchange, and a benchmark NYSE dataset from the US.

The central finding is that besides price impact, waiting time, and the interaction between price impact and waiting time, also matter. The evidence suggests that agents generally do not insist on immediacy, but are willing to wait for a better counter-party to emerge. Price impact is of course still a very important determinant of liquidity. Whether waiting time is also important in explaining liquidity, in addition to price impact, in other exchanges, remains to be examined. Our brief comparison of NSE and NYSE suggests that waiting time is at least also a candidate explanatory variable on other exchanges.

If there is significant non-execution, as we have found on both NSE and NYSE, our work would suggest caution in relying exclusively on price impact measures in analyzing liquidity. This is especially the case when the price impact measures invoke an underlying model such as Kyle (1985) or Glosten and Milgrom (1985) which assumes that all orders are executed.

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TABLE 1. Panel A -- Overall descriptive statistics of some underlying variables

Variable	N	Mean	Std Dev	Min	P1	P10	P25	P50	P75	P90	P99	Max
tau	2905475	40.34	26.46	0.0000	0.00	5.74	17.55	38.04	61.71	78.90	92.36	100.00
age_best	2905475	1.56	19.53	0.0003	0.02	0.10	0.22	0.69	1.79	2.52	4.09	1180.24
age_mean	2905475	3.87	19.75	0.0181	0.94	2.06	2.48	2.83	3.22	3.96	24.00	1180.00
impactcost	2905475	2.47	3.13	0.0003	0.05	0.16	0.35	1.01	3.92	6.58	13.61	230.50
f	2905475	0.10	0.30	0.0000	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00
retIdx	2905475	0.00	0.02	-0.1224	-0.05	-0.02	-0.01	0.00	0.01	0.02	0.05	0.18
retE	2905475	0.00	0.09	-0.9861	-0.11	-0.04	-0.02	0.00	0.01	0.04	0.16	83.60
retEadj	2905475	0.00	0.09	-0.9796	-0.11	-0.04	-0.02	0.00	0.01	0.04	0.15	83.60
fii_buy_Quantity	2905475	34996.54	502291.39	0.0000	0.00	0.00	0.00	0.00	0.00	3937.00	828398.00	342116527.00
fii_buy_value	2905475	10253648.04	125179225.00	0.0000	0.00	0.00	0.00	0.00	0.00	826100.00	232016744.00	73594851590.00
fii_sell_Quantity	2905475	32858.33	376519.99	0.0000	0.00	0.00	0.00	0.00	0.00	1156.00	799732.00	249659659.00
fii_sell_value	2905475	9579182.74	103017993.00	0.0000	0.00	0.00	0.00	0.00	0.00	228157.70	220362897.00	25407658893.00
fii_net_value_scaled	2905475	0.00	0.14	-1.0000	-0.63	0.00	0.00	0.00	0.00	0.00	0.70	1.00

(a) In Panel A above, N is the number of firm-days in the sample.
(b) "Pk" in the column headings refers to the k-th percentile.
(c) Variable definitions are described in detail in Section 3 of the text. Tau = τ, age_best = age in hours of best equity order in the day's order book snapshots, age_mean = age in hours of average equity order in the day's snapshots, impactcost = percent deviation from midpoint of best midquote to execute order of value 100,000, averaged over buy and sell sides, and all snapshots in a day, f = dummy indicating if single-stock futures exist, retIdx = daily return on NIFTY index (counted several times each day in this table, once for each firm), retE = daily firm return, retEadj = retE – retIdx (market-adjusted firm return), fii_buy(sell)_quantity = number of shares in FII buy (sell) orders executed during the day, fii_net_value_scaled = net cash inflow by FIIs, scaled by daily trading value.

TABLE 1. Panel B -- Firm-by-firm descriptive statistics of some underlying variables

Variable	N	Mean	Std Dev	Min	P1	P10	P25	P50	P75	P90	P99	Max
tau	2607	35.70	18.61	0.0000	5.02	13.35	21.23	32.93	47.33	62.57	83.37	97.54
age_best	2607	341950.67	1215056.47	0.0000	120.00	1638.78	6164.68	36511.01	160736.31	674212.22	5718041.38	22413019.69
age_mean	2607	1.68	1.70	0.0726	0.23	0.43	0.74	1.42	2.20	3.00	6.54	51.69
impactcost	2607	4.22	2.56	0.5362	2.20	2.69	2.91	3.46	4.39	6.90	13.97	58.77
f	2607	3.23	2.89	0.0554	0.15	0.34	0.79	2.36	4.95	7.66	11.45	13.62
retIdx	2607	0.06	0.19	0.0000	0.00	0.00	0.00	0.00	0.00	0.26	0.98	1.00
retE	2607	0.00	0.00	-0.0520	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01
retEadj	2607	0.00	0.01	-0.0936	-0.01	0.00	0.00	0.00	0.00	0.01	0.03	0.15
fii_buy_Quantity	2607	0.00	0.01	-0.0904	-0.01	0.00	0.00	0.00	0.00	0.01	0.03	0.14
fii_buy_value	2607	29551.26	172144.84	0.0000	0.00	0.00	0.00	0.00	2985.45	37185.61	693717.96	4543402.16
fii_sell_Quantity	2607	7240360.23	49124011.89	0.0000	0.00	0.00	0.00	0.00	455227.10	7183272.94	162601200.00	1472556223.00
fii_sell_value	2607	26889.68	138488.54	0.0000	0.00	0.00	0.00	0.00	3970.03	33054.23	638991.51	2673692.02
fii_net_value_scaled	2607	6574554.01	42559734.09	0.0000	0.00	0.00	0.00	0.00	449775.53	6148968.65	151513781.00	802188320.00

(a) In the firm-by-firm descriptive statistics shown in Panel B above, we first compute the firm-specific mean for each of the variables. So τ (tau) really means mean τ in Panel B. Then we compute descriptive statistics across firms. So N in Panel B is the number of firms.
(b) "Pk" in the column headings refers to the k-th percentile.
(c) Variable definitions are described in detail in Section 3 of the text. Tau = τ, age_best = age in hours of best equity order in the day's order book snapshots, age_mean = age in hours of average equity order in the day's snapshots, impactcost = percent deviation from midpoint of best midquote to execute order of value 100,000, averaged over buy and sell sides, and all snapshots in a day, f = dummy indicating if single-stock futures exist, retIdx = daily return on NIFTY index (counted several times each day in this table, once for each firm), retE = daily firm return, retEadj = retE – retIdx (market-adjusted firm return), fii_buy(sell)_quantity = number of shares in FII buy (sell) orders executed during the day, fii_net_value_scaled = net cash inflow by FIIs, scaled by daily trading value.

TABLE 2, Panel A – Descriptive Statistics of τ (Tau) By Activity Level

TAU	N	Mean	Std Dev	Min	P1	P25	P50	P75	Max
Overall	2905475	40.34	26.46	0.00	0.00	17.55	38.04	61.71	100.00
BY QUINTILE									
1 (highest vol.)	582403	63.83	22.29	0.00	6.03	49.42	68.85	82.00	100.00
2	579725	48.39	23.10	0.00	1.26	30.58	49.05	66.83	99.99
3	580751	37.76	22.62	0.00	0.00	19.42	35.99	54.38	99.92
4	582334	30.47	21.89	0.00	0.00	12.44	26.81	45.31	99.97
5 (lowest vol.)	580262	21.21	19.80	0.00	0.00	5.10	15.82	32.35	99.96

Quintiles above were formed by first computing average daily volume for each firm, and then grouping firms by quintile ranks formed based on average daily volume (1 = largest daily volume, 5 = least average daily volume). Only the grouping of firms uses the average daily volume. The descriptive statistics themselves use the raw τ numbers, not firm-specific averages.

TABLE 2, Panel B – Descriptive Statistics of τ (Tau) By Time

TAU	N	Mean	Std Dev	Min	P1	P25	P50	P75	Max
Overall	2905475	40.34	26.46	0.00	0.00	17.55	38.04	61.71	100.00
1999	284577	33.32	24.79	0.00	0.00	12.20	29.54	51.30	99.91
2000	241221	31.54	26.30	0.00	0.00	8.57	25.66	50.32	99.93
2001	224170	25.92	25.21	0.00	0.00	4.51	18.18	41.07	99.92
2002	190437	30.85	23.59	0.00	0.00	11.85	25.72	45.35	99.91
2003	174361	41.38	25.40	0.00	0.00	20.71	39.37	60.65	99.94
2004	186382	45.42	24.55	0.00	0.00	26.22	45.12	64.40	99.45
2005	183334	52.34	21.87	0.00	4.04	36.23	53.17	69.71	99.46
2006	208526	50.82	23.74	0.00	2.32	32.40	51.42	70.41	99.73
2007	256576	48.09	25.26	0.00	0.79	27.58	48.00	69.41	99.95
2008	273079	44.68	27.40	0.00	0.07	20.76	43.32	68.35	100.00
2009	271911	45.03	27.27	0.00	0.02	20.97	44.44	68.65	100.00
2010	324821	38.97	24.99	0.00	0.89	17.49	35.35	58.85	99.80
2011	86080	36.11	26.62	0.00	0.13	12.79	30.50	57.48	99.69

"Pk" in the column headings refers to the k-th percentile.

Table 3. Regression of τ (Tau) on various determinants

AGGREGATE SAMPLE	PRICE	TIME	P*T	F	F*Price	F*Time	FII	FII*Price	FII*Time	Adj Ret		Adj R^2
Model 1 coefficients	-6.389	-4.003	0.010									0.340
t	-376.300	-588.220	17.930									
Model 2 coefficients	-5.270	-3.529	0.008	24.211	0.025	0.497						0.411
t	-323.990	-543.650	17.920	273.490	14.460	1.690						
Model 3 coefficients	-5.270	-3.529	0.008	24.198	0.025	0.514	0.759	-0.004	-0.493			0.411
t	-323.980	-543.640	17.920	273.430	14.320	1.750	4.760	-0.550	-1.110			
Model 4 coefficients	-5.271	-3.532	0.008	24.211	0.025	0.429	0.818	-0.004	-0.552	-25.085		0.411
t	-324.140	-543.940	17.910	271.950	14.260	1.440	5.120	-0.560	-1.240	-32.170		

1. The sample variables were winsorized at 2% before being used in the regression analyses. Also, since FII data is available only since 2002, though FII activity in India existed even before that, for all the regressions, the time period for the sample starts with 2002.
2. All the regression models were run with the intercept, and used Generalized Method of Moments with Newey-West standard errors.
3. Variable definitions (more details in the text) for Models 1-4 (LHS = τ)
   PRICE = Impact-cost percent
   TIME = Age_of_the_Best_Order
   F = Futures Dummy (1=yes, futures exist)
   FII = (FII_Val_B-FII_Val_S)/Daily_Trading_Value (net cash inflow due to Foreign Institutional Investors, scaled by Daily Trading Value, same as fii_net_value_scaled in Table 1)
   Adj_Ret = Firm Return Adjusted by Market Return (same as retEadj in Table 1)

TABLE 4. SUMMARY of Firm-by-Firm Regression Analyses

Model 1	PRICE	TIME	P*T	F	F*Price	F*Time	FII	FII*Price	FII*Time	Adj Ret		Adj R^2
Mean	4.45	1.87	3.69									0.136
Std dev	401.47	175.80	248.62									0.151
Min	-3780.28	-1125.08	-3340.59									-2.889
P10	-9.49	-7.14	-0.04									0.015
P25	-6.84	-4.35	0.00									0.041
P50	-4.71	-2.66	0.06									0.098
P75	-2.99	-0.80	0.76									0.203
P90	-1.16	1.97	3.07									0.325
Max	16801.40	7294.32	11825.76									0.896

Model 2	PRICE	TIME	P*T	F	F*Price	F*Time	FII	FII*Price	FII*Time	Adj Ret		Adj R^2
Mean	0.33	0.96	5.50	0.54	2.52	0.86						0.131
Std dev	438.61	164.84	274.23	4.04	212.53	25.94						0.149
Min	-8751.58	-1125.08	-3340.59	-34.32	-3989.34	-693.99						-2.889
P10	-10.54	-10.37	-0.45	0.00	0.00	0.00						0.019
P25	-7.27	-5.22	0.00	0.00	0.00	0.00						0.046
P50	-4.93	-3.03	0.12	0.00	0.00	0.00						0.100
P75	-3.02	-1.08	1.41	0.00	0.00	0.00						0.196
P90	-0.72	3.72	4.29	0.00	0.00	3.81						0.301
Max	16801.40	7294.32	11825.76	29.33	8704.27	589.63						0.896

(a) Table 4 summarizes results for which firm-by-firm regressions were run , for 2607, 2086, 2003, and 2003 firms respectively, for Models 1, 2, 3 and 4.
(b) "Pk" in the column headings refers to the k-th percentile
(c) Variable definitions (more details in the text) for Models 1-4 (LHS = τ)

PRICE = Impact-cost percent
TIME = Age_of_the_Best_Order
F = Futures Dummy (1=yes, futures exist)
FII = (FII_Val_B-FII_Val_S)/Daily_Trading_Value (net cash inflow due to Foreign Institutional Investors, scaled by Daily Trading Value, same as fii_net_value_scaled in Table 1)
Adj_Ret = Firm Return Adjusted by Market Return (same as retEadj in Table 1)

TABLE 4. SUMMARY of Firm-by-Firm Regression Analyses (continued)

Model 3	PRICE	TIME	P*T	F	F*Price	F*Time	FII	FII*Price	FII*Time	Adj Ret	Adj R^2
Mean	-0.46	0.83	2.65	0.56	3.23	0.91	691.34	595.10	-390.56		0.126
Std dev	451.98	169.52	308.01	3.99	217.08	21.95	106533.42	183273.84	14858.05		0.137
Min	-8751.58	-1125.08	-5700.73	-34.31	-3948.37	-477.15	-2132540.41	-2789952.77	-654473.55		-2.889
P10	-10.32	-9.99	-0.50	0.00	0.00	0.00	-12.91	-8.49	-13.36		0.021
P25	-7.22	-5.16	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.046
P50	-4.84	-2.99	0.10	0.00	0.00	0.00	0.00	0.00	0.00		0.095
P75	-2.85	-0.98	1.39	0.00	0.00	0.00	0.85	0.00	0.00		0.184
P90	-0.55	4.28	4.39	0.91	0.05	3.43	18.90	7.50	10.78		0.280
Max	16801.40	7294.32	11825.76	29.48	8704.27	568.34	3471757.51	7258030.46	45012.96		0.896

Model 4	PRICE	TIME	P*T	F	F*Price	F*Time	FII	FII*Price	FII*Time	Adj Ret	Adj R^2
Mean	-1.09	1.01	1.94	0.57	3.91	0.91	646.04	599.30	-389.22	-29.33	0.129
Std dev	440.34	172.30	311.39	4.00	213.48	22.20	105325.15	184398.68	14940.68	144.65	0.118
Min	-9021.46	-1125.08	-5700.73	-34.26	-2807.78	-474.37	-2134625.86	-2718393.27	-658902.12	-3847.30	-0.688
P10	-10.33	-10.17	-0.45	0.00	0.00	0.00	-12.86	-8.65	-13.76	-85.61	0.021
P25	-7.21	-5.22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-55.98	0.047
P50	-4.85	-3.02	0.10	0.00	0.00	0.00	0.00	0.00	0.00	-26.97	0.097
P75	-2.86	-1.04	1.42	0.00	0.00	0.00	0.92	0.00	0.00	0.92	0.185
P90	-0.62	4.22	4.46	0.86	0.05	3.46	18.80	7.43	10.98	40.79	0.281
Max	16222.90	7408.86	11825.76	29.41	8971.33	564.35	3382652.17	7313026.65	44733.19	1075.51	0.900

(a) Table 4 summarizes results for which firm-by-firm regressions were run , for 2607, 2086, 2003, and 2003 firms respectively, for Models 1, 2, 3 and 4.

(b) "Pk" in the column headings refers to the k-th percentile

(c) Variable definitions (more details in the text) for Models 1-4 (LHS = τ)

PRICE = Impact-cost percent

TIME = Age_of_the_Best_Order

F = Futures Dummy (1=yes, futures exist)

FII = (FII_Val_B-FII_Val_S)/Daily_Trading_Value (net cash inflow due to Foreign Institutional Investors, scaled by Daily Trading Value, same as fii_net_value_scaled in Table 1)

Adj_Ret = Firm Return Adjusted by Market Return (same as retEadj in Table 1)

Table 5. Relationship between price impact (PRICE) and τ

Panel A -- dTau/dPRICE, evaluated at each observed value of the interaction variables; The variable of analysis denotes the proportion of each firm's observations where dELS/dPRICE > 0

dTau/dPRICE	N (# firms)	Mean	Std Dev	Min	P1	P10	P25	P50	P75	P90	P99	Max
Model 1	2607	0.20	0.37	0	0	0	0	0.00	0.14	1.00	1	1
Model 2	2086	0.23	0.36	0	0	0	0	0.00	0.34	1.00	1	1
Model 3	2003	0.24	0.36	0	0	0	0	0.01	0.34	1.00	1	1
Model 4	2003	0.23	0.36	0	0	0	0	0.01	0.34	1.00	1	1

Panel B — dTau/dPRICE, evaluated at each firm's mean value of the interaction variables

dTau/dPRICE	N (# firms)	Mean	Std Dev	Min	P1	P10	P25	P50	P75	P90	P99	Max
Model 1	2607	-1.69	30.43	-383.45	-20.39	-6.22	-3.94	-2.38	-0.53	2.27	18.31	1419.32
Model 2	2086	-1.20	33.69	-240.42	-21.55	-7.41	-4.31	-2.51	-0.44	4.22	24.32	1419.32
Model 3	2003	-0.91	41.36	-388.11	-21.98	-7.09	-4.27	-2.52	-0.35	4.46	22.97	1419.32
Model 4	2003	-0.85	47.68	-388.11	-22.38	-7.18	-4.31	-2.53	-0.36	4.48	23.00	1743.56

Panel C — dTau/dPRICE, evaluated at each firm's median value of the interaction variables

dTau/dPRICE	N (# firms)	Mean	Std Dev	Min	P1	P10	P25	P50	P75	P90	P99	Max
Model 1	2607	-4.52	22.98	-351.24	-15.13	-8.50	-6.22	-4.49	-2.89	-1.19	3.76	992.11
Model 2	2086	-4.36	26.09	-379.14	-16.28	-8.92	-6.40	-4.51	-2.65	-0.57	6.01	992.11
Model 3	2003	-5.95	56.77	-1537.87	-23.91	-8.95	-6.42	-4.36	-2.56	-0.61	6.58	992.11
Model 4	2003	-5.95	57.12	-1537.87	-27.81	-8.97	-6.44	-4.38	-2.58	-0.63	7.09	992.11

(a) In Panel A the variable of analysis is the proportion of each firm's obervations with dTau/dPRICE>0. So we first get one number for each firm-day, which reduces to one number per firm once we take the proportion.

(b) In Panels B and C the variable of analysis is dTau/dPRICE evaluated at one point, so we start with only one number per firm.

TABLE 6 -- Simultaneous equation model for τ (Tau) and Firm Returns

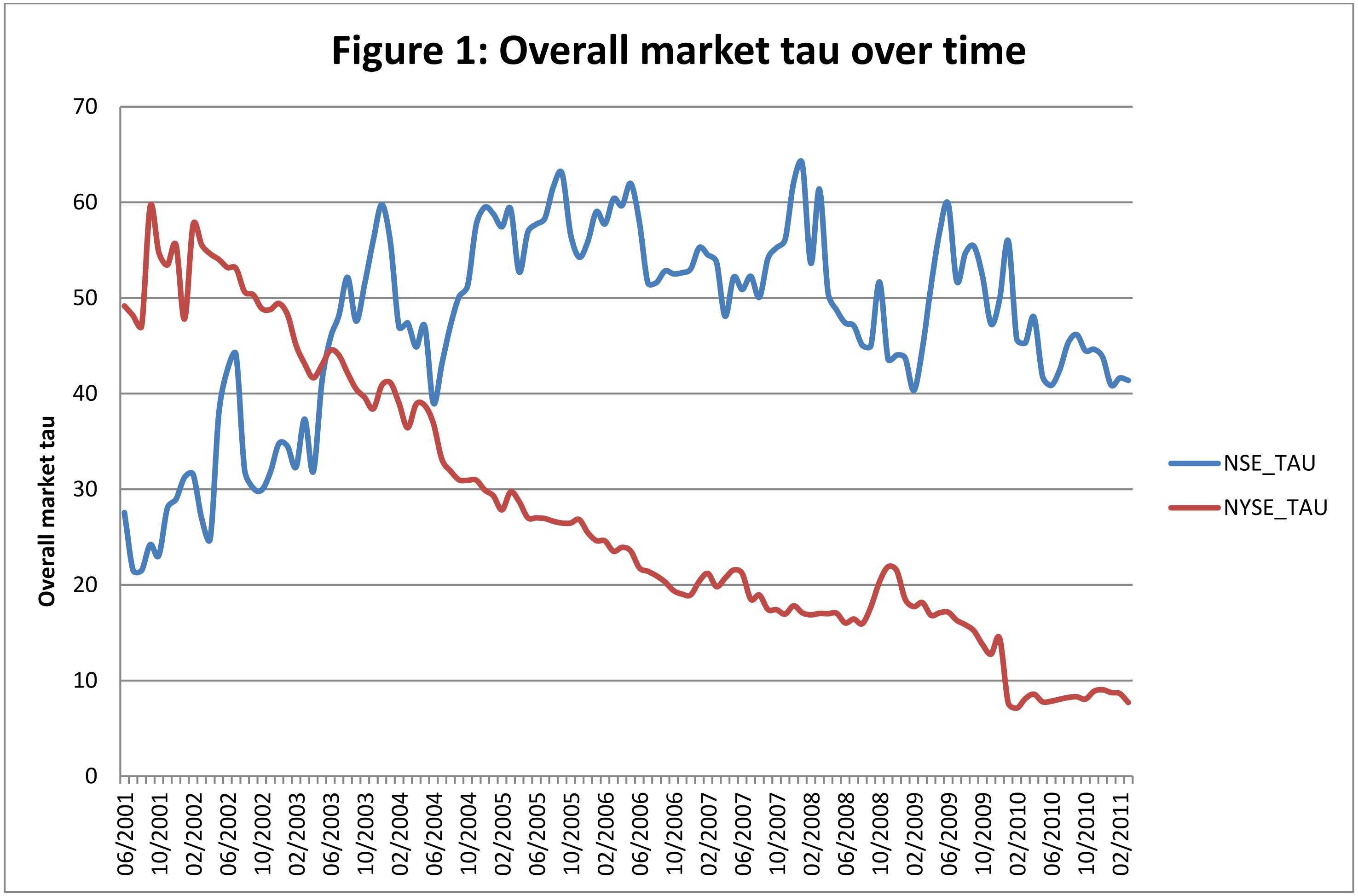
Parameter	Estimate	Std Err	t Value	Approx Pr > |t|	Parameter	Estimate	Std Err	t Value	Approx Pr > |t|
a0 (intercept, tau equation)	19.0635	0.032	590.25	<.0001	b0 (intercept, firm-return equation)	-0.0055	0.000	-75.7	<.0001
a1 (Firm-return)	92.4816	0.333	277.78	<.0001	b1 (Tau)	0.0004	0.000	277.78	<.0001
a2 (lag tau)	0.6343	0.001	1190.76	<.0001	b2 (lag tau)	-0.0003	0.000	-184.6	<.0001
a3 (lag Firm-return)	-1.0162	0.113	-9.01	<.0001	b3 (lag Firm-return)	0.0013	0.000	5.71	<.0001
a4 (TIME == age of best order)	-2.2814	0.013	-171.93	<.0001	b4 (TIME == age of best order)	-0.0002	0.000	-8.4	<.0001
a5 (PRICE == impact cost)	-1.2861	0.006	-219.47	<.0001	b5 (PRICE == impact cost)	0.0002	0.000	19.13	<.0001
a6 (TIME*PRICE)	0.0019	0.000	13.74	<.0001	b6 (TIME*PRICE)	0.0000	0.000	-2.59	0.0095
a7 (F == Futures Dummy)	8.7408	0.043	205.52	<.0001	b7 (F == Futures Dummy)	-0.0042	0.000	-46.34	<.0001
a8 (TIME*F)	0.0098	0.002	5.3	<.0001	b8 (TIME*F)	0.0000	0.000	-0.56	0.5787
a9 (PRICE*F)	1.2463	0.095	13.14	<.0001	b9 (PRICE*F)	-0.0015	0.000	-7.68	<.0001
a10 (lag FII)	-0.0844	0.080	-1.05	0.2915	b10 (lag FII)	0.0000	0.000	0.11	0.9105
a11 (TIME*lag FII)	-0.0067	0.005	-1.43	0.1532	b11 (TIME*lag FII)	0.0000	0.000	0.46	0.6479
a12 (PRICE*lag FII)	-0.3245	0.181	-1.79	0.0732	b12 (PRICE*lag FII)	-0.0020	0.000	-5.19	<.0001
Adj R-Sq	0.6357				Adj R-Sq	0.0388			

Simultaneous equation model summary:

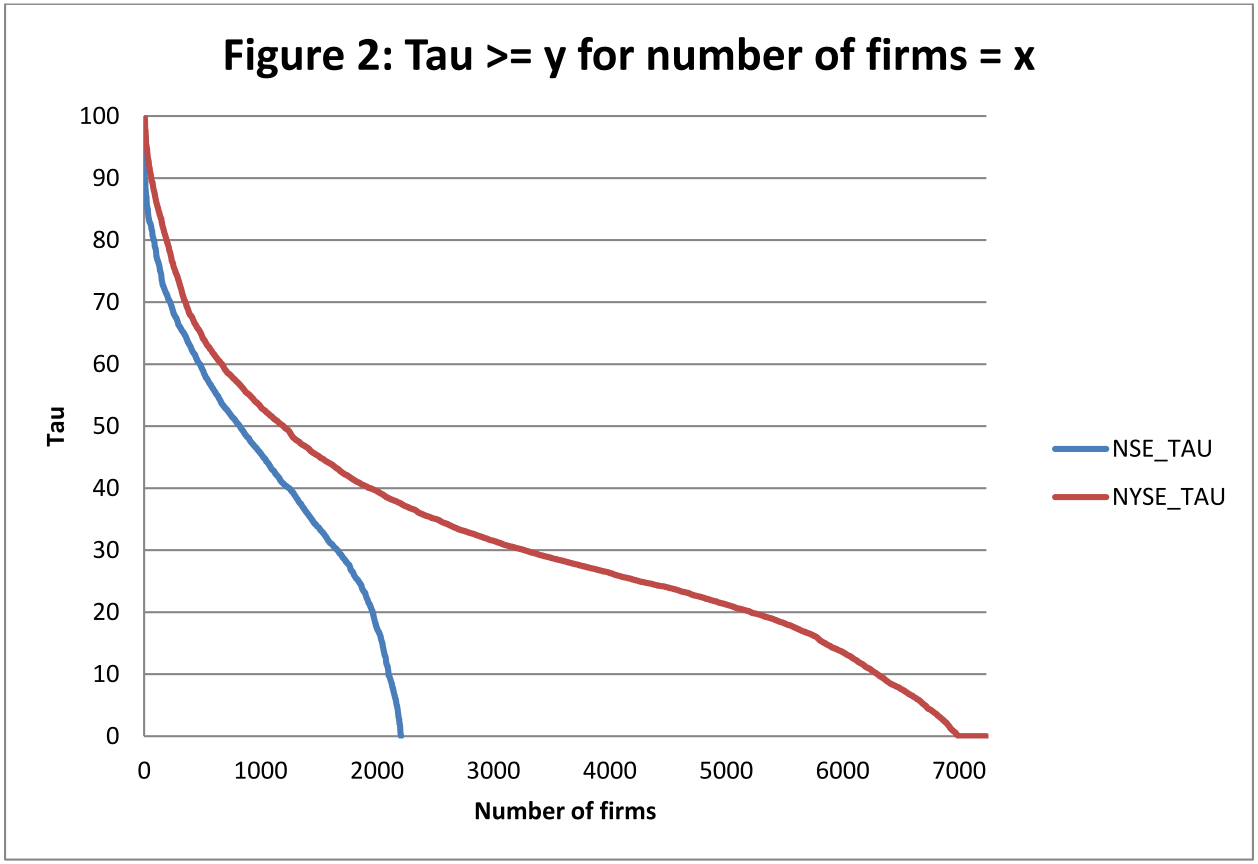
Tau = a0+a1*(Firm-return)+a2*(lag_Tau)+a3*(lag_Firm-return)+a4*(TIME)+a5*(PRICE)+a6*(TIME*PRICE) +a7*(F)+a8*(TIME*F)+a9*(PRICE*F)+a10*(lag_FII)+a11*(TIME*lag_FII)+a12*(PRICE*lag_FII)+e_tau

RetEadj = b0+b1*(Tau)+b2*(lag_Tau)+b3*(lag_Firm-return)+b4*(TIME)+b5*(PRICE)+b6*(TIME*PRICE) +b7*(F)+b8*(TIME*F)+b9*(PRICE*F)+b10*(lag_ FII)+b11*(TIME*lag_ FII)+b12*(PRICE*lag_ FII)+e_ret

where TIME = Age_of_the_Best_Order, PRICE = Impact-cost percent, F = Futures Dummy (1=yes, futures exist), FII = (FII_Val_B-FII_Val_S)/Daily_Trading_Value (net cash inflow due to Foreign Institutional Investors, scaled by Daily Trading Value, same as fii_net_value_scaled in Table 1), Adj_Ret = Firm Return Adjusted by Market Return (same as retEadj in Table 1).



During our comparison period for this plot, June 2001 to March 2011, NSE had 2208 firms; NYSE, 7235 firms.



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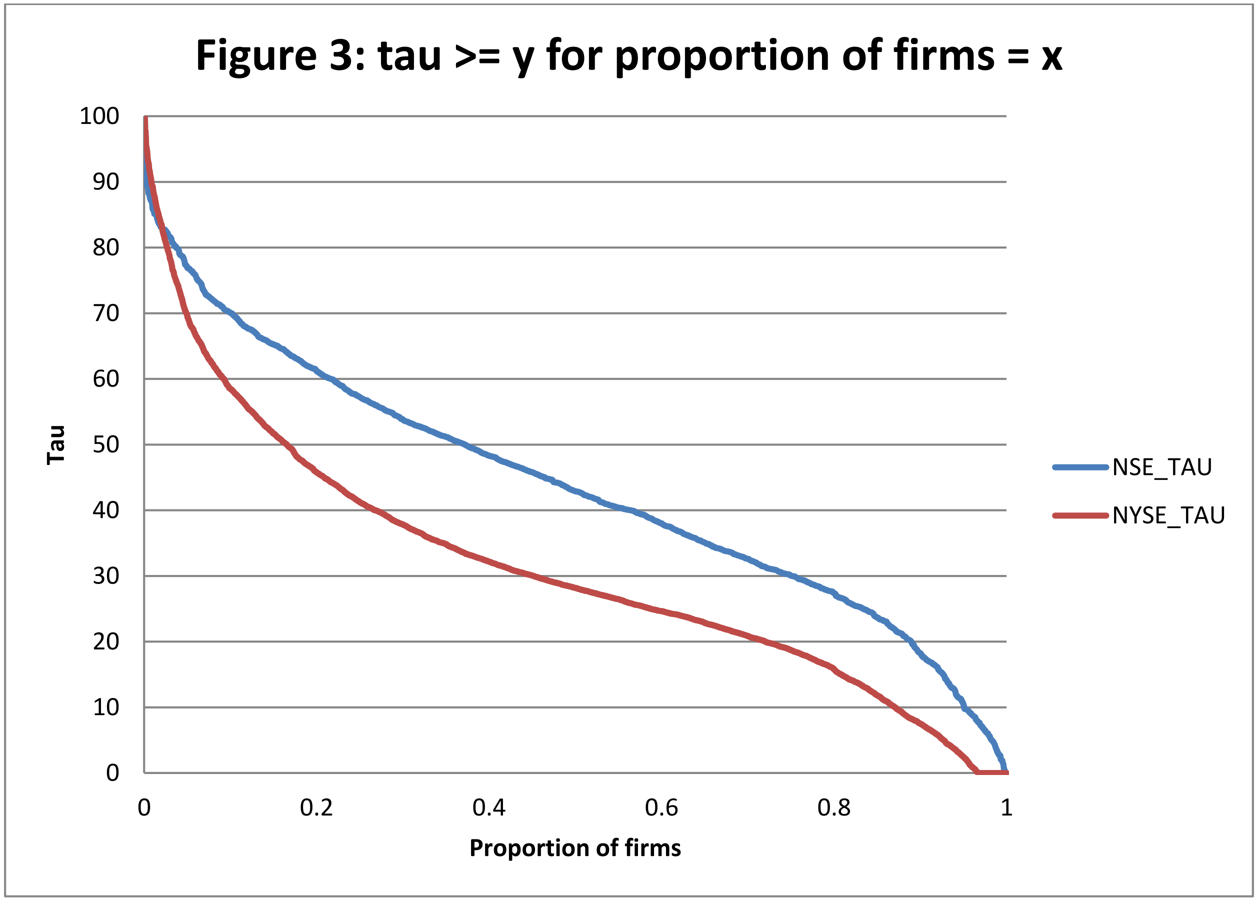

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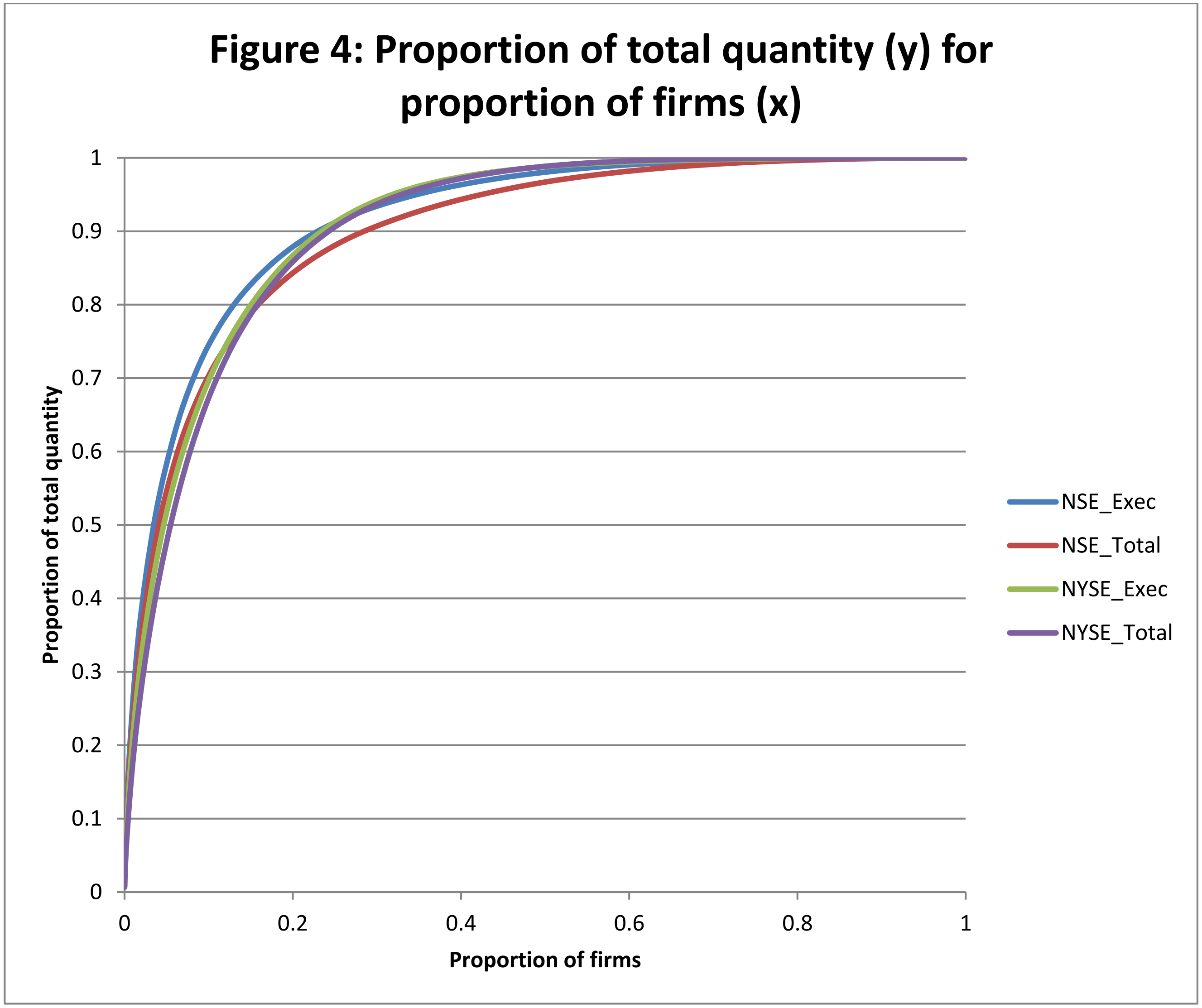
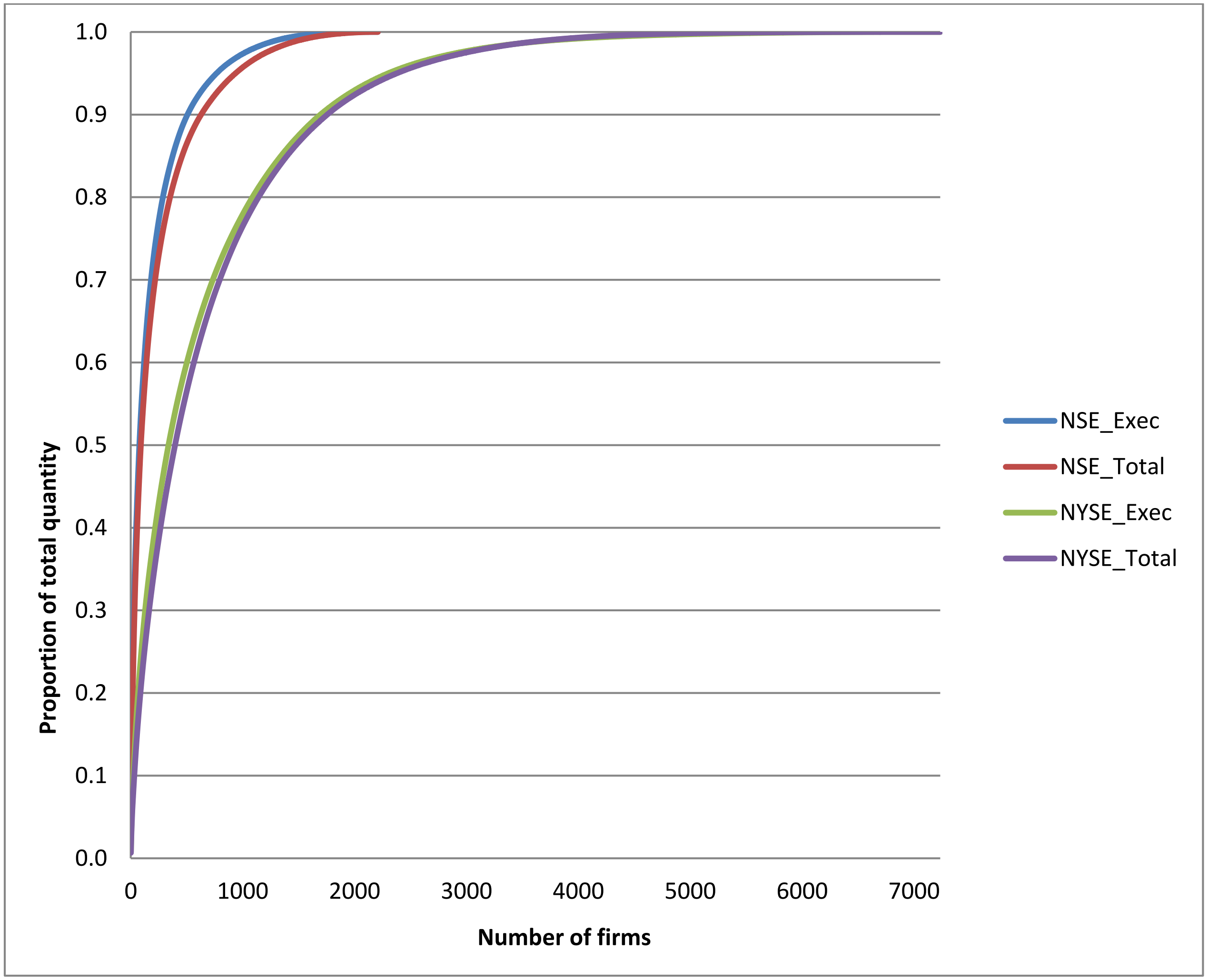


"_Exec" = quantity in orders that are executed; "_Total" = quantity in total orders received. During our comparison period for this plot, June 2001 to March 2011, NSE had 2208 firms; NYSE, 7235 firms.

Figure 5: Proportion of total quantity (y) for number of firms (x)



"_Exec" = quantity in orders that are executed; "_Total" = quantity in total orders received. During our comparison period for this plot, June 2001 to March 2011, NSE had 2208 firms; NYSE, 7235 firms.

Murugappa (Murgie) Krishnan

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